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Subject: Mathematics
Class: Class X (CBSE)

PERIODIC TEST – 02 (PROBABILITY)
Chapter 14: Probability

Time: 60 Minutes
Max. Marks: 30

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

In a family of 3 children, the probability of having at least one boy is:

(a) $\frac{7}{8}$ (b) $\frac{1}{8}$

(c) $\frac{5}{8}$ (d) $\frac{3}{4}$

Working & Calculation:

- In a family of 3 children, total possibilities $n(S) = 2^3 = 8$:

$$S = \{\text{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG}\}$$

- The event "at least one boy" includes all outcomes except GGG (3 girls):

$$P(\text{at least one boy}) = 1 - P(\text{all girls}) = 1 - \frac{1}{8} = \frac{7}{8}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{7}{8}$

Question 2 [1 Mark]

Two dice are thrown simultaneously. What is the probability of getting a sum greater than 9?

(a) $\frac{1}{6}$ (b) $\frac{5}{36}$

(c) $\frac{1}{9}$ (d) $\frac{1}{12}$

Working & Calculation:

- Total outcomes on rolling two dice: $n(S) = 6 \times 6 = 36$.
- Sums strictly greater than 9 are sums equal to 10, 11, or 12:

$$E = \{(4,6),(5,5),(6,4),(5,6),(6,5),(6,6)\} \implies n(E) = 6$$

- Therefore:

$$P(\text{sum } > 9) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{6}$

Question 3 [1 Mark]

A card is drawn from a well-shuffled pack of 52 cards. The probability that the card is a spade or an ace is:

(a) $\frac{4}{13}$ (b) $\frac{17}{52}$

(c) $\frac{1}{4}$ (d) $\frac{2}{13}$

Working & Calculation:

- Total cards in the deck: $n(S) = 52$.
- Number of spades $= 13$. Number of aces in remaining suits $= 3$ (Ace of spades is already counted).
- Favourable cards $n(E) = 13 + 3 = 16 \implies P(\text{spade or ace}) = \frac{16}{52} = \frac{4}{13}$.

CORRECT ANSWER: OPTION (a) ⟶ $\frac{4}{13}$

Question 4 [1 Mark]

A bag contains 6 red, 4 white, and 8 blue balls. If one ball is drawn, the probability that the drawn ball is not red is:

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$

(c) $\frac{4}{9}$ (d) $\frac{5}{9}$

Working & Calculation:

- Total balls in the bag: $n(S) = 6 + 4 + 8 = 18$.
- Balls that are not red (white + blue): $4 + 8 = 12$ balls.

$$P(\text{not red}) = \frac{12}{18} = \frac{2}{3} \quad \left[\text{or via complement: } 1 - \frac{6}{18} = 1 - \frac{1}{3} = \frac{2}{3}\right]$$

CORRECT ANSWER: OPTION (a) ⟶ $\frac{2}{3}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): If all red face cards are removed from a pack of 52 cards, the probability of drawing an ace from the remaining cards is $\frac{2}{23}$.

Reason (R): After removing 6 red face cards, 46 cards remain in the deck, including all 4 aces.

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Options:

(a) **Both (A) and (R) are true, and (R) is the correct explanation of (A).**
(b) **Both (A) and (R) are true, but (R) is not the correct explanation of (A).**
(c) **(A) is true, but (R) is false.**
(d) **(A) is false, but (R) is true.**

Step-by-Step Logical Analysis:

1. **Reason (R):** Red face cards removed $= 2 \times 3 = 6$ cards. Remaining cards $= 52 - 6 = 46$. Since aces are not face cards, all 4 aces remain. $\implies$ **(R) is TRUE.**
2. **Assertion (A):** Favourable aces $= 4$, total remaining $= 46 \implies P = \frac{4}{46} = \frac{2}{23}$. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** (R) contains the exact card counts that justify (A). Hence, (R) correctly explains (A).

CORRECT ANSWER: OPTION (a) ⟶ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A die is thrown once. Find the probability of getting: (i) a number lying strictly between 2 and 6 (ii) an odd prime number.

Sample Space: $S = \{1, 2, 3, 4, 5, 6\} \implies n(S) = 6$.

(i) Lying strictly between 2 and 6 [1 Mark]

Numbers strictly between 2 and 6: $E_1 = \{3, 4, 5\} \implies n(E_1) = 3 \implies P = \frac{3}{6} = \frac{1}{2}$.

Answer (i): $\quad \mathbf{P(\text{between 2 and 6}) = \frac{1}{2}}$

(ii) An odd prime number [1 Mark]

Prime numbers are $\{2, 3, 5\}$. Odd primes are $E_2 = \{3, 5\} \implies n(E_2) = 2 \implies P = \frac{2}{6} = \frac{1}{3}$.

Answer (ii): $\quad \mathbf{P(\text{odd prime}) = \frac{1}{3}}$

Question 7 [2 Marks]

A lot consists of 144 ball pens of which 20 are defective and the others are good. A customer buys a pen selected at random. What is the probability that: (i) she will buy it (if it is good)? (ii) she will not buy it (if it is defective)?

Given Data: Total pens $n(S) = 144$, defective $= 20$, good pens $= 144 - 20 = 124$.

(i) She will buy it (pen is good) [1 Mark]

Favourable outcomes $= 124 \implies P(\text{buys pen}) = \frac{124}{144} = \frac{31}{36}$.

Answer (i): $\quad \mathbf{P(\text{she will buy it}) = \frac{31}{36}}$

(ii) She will not buy it (pen is defective) [1 Mark]

Favourable outcomes $= 20 \implies P(\text{will not buy}) = \frac{20}{144} = \frac{5}{36}$.

Answer (ii): $\quad \mathbf{P(\text{she will not buy it}) = \frac{5}{36}}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

Find the probability that:
(i) a leap year has 53 Sundays and 53 Mondays.
(ii) a leap year has 53 Sundays or 53 Mondays.
(iii) a non-leap year has 53 Sundays.

Leap Year Setup: 366 days = 52 weeks + 2 extra days (7 pairs: Sun–Mon, Mon–Tue, Tue–Wed, Wed–Thu, Thu–Fri, Fri–Sat, Sat–Sun).

(i) 53 Sundays AND 53 Mondays in a leap year [1 Mark]

Favourable outcome is the single pair $\{(\text{Sun, Mon})\} \implies P = \frac{1}{7}$.

Answer (i): $\quad \mathbf{P(\text{53 Sundays and 53 Mondays}) = \frac{1}{7}}$

(ii) 53 Sundays OR 53 Mondays in a leap year [1 Mark]

Favourable pairs containing Sunday or Monday: $\{(\text{Sat, Sun}), (\text{Sun, Mon}), (\text{Mon, Tue})\} \implies P = \frac{3}{7}$.

Answer (ii): $\quad \mathbf{P(\text{53 Sundays or 53 Mondays}) = \frac{3}{7}}$

(iii) 53 Sundays in a non-leap year [1 Mark]

A non-leap year has 1 extra day (7 choices). Sunday is 1 choice $\implies P = \frac{1}{7}$.

Answer (iii): $\quad \mathbf{P(\text{53 Sundays in non-leap year}) = \frac{1}{7}}$

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Question 9 [3 Marks]

From the letters of the word 'MATHEMATICS', a letter is chosen at random. Find the probability that the letter chosen is: (i) a vowel (ii) the letter 'M' (iii) a consonant.

Sample Space: Total letters in 'MATHEMATICS': $n(S) = 11$.
Breakdown: M (2), A (2), T (2), H (1), E (1), I (1), C (1), S (1).

(i) A vowel [1 Mark]

Vowels are $\{A, E, A, I\} \implies n(\text{vowels}) = 4 \implies P(\text{vowel}) = \frac{4}{11}$.

Answer (i): $\mathbf{P(\text{a vowel}) = \frac{4}{11}}$

(ii) The letter 'M' [1 Mark]

Number of 'M's $= 2 \implies P(\text{letter 'M'}) = \frac{2}{11}$.

Answer (ii): $\mathbf{P(\text{letter 'M'}) = \frac{2}{11}}$

(iii) A consonant [1 Mark]

Consonants $= 11 - 4 = 7 \implies P(\text{consonant}) = \frac{7}{11}$.

Answer (iii): $\mathbf{P(\text{a consonant}) = \frac{7}{11}}$

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

Question 10 [5 Marks]

From a pack of 52 playing cards, all face cards (Kings, Queens, Jacks) and aces are removed. The remaining cards are well shuffled and then one card is drawn at random. Find the probability of getting:
(i) a black card (ii) a red card of value 8 (iii) a number card which is a multiple of 3
(iv) a card with a prime number (v) a card with an even number.

Remaining Card Set: Cards removed: 12 face cards + 4 aces = 16 cards.
Remaining cards $n(S') = 52 - 16 = 36$ cards, bearing numbers $\{2, 3, 4, 5, 6, 7, 8, 9, 10\}$ across 4 suits $(9 \times 4 = 36)$.

(i) A black card: Black suits (Spades, Clubs) have $9 + 9 = 18$ cards $\implies P = \frac{18}{36} = \frac{1}{2}$.

Answer (i): $\mathbf{P(\text{black card}) = \frac{1}{2}}$

(ii) A red card of value 8: Red suits (Hearts, Diamonds) contain 2 eights $\implies P = \frac{2}{36} = \frac{1}{18}$.

Answer (ii): $\mathbf{P(\text{red } 8) = \frac{1}{18}}$

(iii) A number card multiple of 3: Multiples of 3 in $\{2, \ldots, 10\}$ are $\{3, 6, 9\}$ $(3 \times 4 = 12 \text{ cards}) \implies P = \frac{12}{36} = \frac{1}{3}$.

Answer (iii): $\mathbf{P(\text{multiple of } 3) = \frac{1}{3}}$

(iv) A card with a prime number: Primes in $\{2, \ldots, 10\}$ are $\{2, 3, 5, 7\}$ $(4 \times 4 = 16 \text{ cards}) \implies P = \frac{16}{36} = \frac{4}{9}$.

Answer (iv): $\mathbf{P(\text{prime number}) = \frac{4}{9}}$

(v) A card with an even number: Even numbers in $\{2, \ldots, 10\}$ are $\{2, 4, 6, 8, 10\}$ $(5 \times 4 = 20 \text{ cards}) \implies P = \frac{20}{36} = \frac{5}{9}$.

Answer (v): $\mathbf{P(\text{even number}) = \frac{5}{9}}$

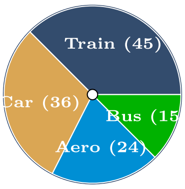
SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11 (Case Study Analysis) [4 Marks]

An environmental NGO surveyed 120 residents of a locality to study transportation choices made during their last vacation trip. The gathered data revealed:

· Train: 45 travellers · Personal Car: 36 travellers
· Aeroplane: 24 travellers · Public Bus: 15 travellers

A resident is chosen at random from this group. Based on the above scenario, answer the following:



Sample Space: Total surveyed travellers $n(S) = 45 + 36 + 24 + 15 = 120$.

(i) Probability that the chosen resident travelled by car [1 Mark]

Number of car travellers = 36:

$$P(\text{car}) = \frac{36}{120} = \frac{3}{10} = 0.3$$

Answer (i): $\mathbf{P(\text{travelled by car}) = \frac{3}{10} = 0.3}$

(ii) Probability that resident travelled by public mass transit (train or bus) [1 Mark]

Favourable travellers = Train (45) + Bus (15) = 60:

$$P(\text{mass transit}) = \frac{60}{120} = \frac{1}{2} = 0.5$$

Answer (ii): $\mathbf{P(\text{public mass transit}) = \frac{1}{2} = 0.5}$

(iii) Probability that the resident did NOT travel by aeroplane [1 Mark]

Travellers who did not take an aeroplane: $120 - 24 = 96$:

$$P(\text{not aeroplane}) = \frac{96}{120} = \frac{4}{5} = 0.8 \quad \left[\text{or via complement: } 1 - \frac{24}{120} = 1 - 0.2 = 0.8\right]$$

Answer (iii): $\mathbf{P(\text{not by aeroplane}) = \frac{4}{5} = 0.8}$

(iv) Central angle corresponding to aeroplane travellers in a pie chart [1 Mark]

The total central angle of a complete circle is 360°:

$$\text{Central Angle} = \frac{\text{Aeroplane Travellers}}{\text{Total Travellers}} \times 360^\circ = \frac{24}{120} \times 360^\circ = \frac{1}{5} \times 360^\circ = 72^\circ$$

Answer (iv): $\mathbf{\text{Central Angle} = 72^\circ}$

— END OF WORKED SOLUTIONS (TEST–08) · MATHS MANTHAN TUITION CLASSES —

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